UNDER PRESSURE

Predicting the players coaches can trust

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THE FINAL STRETCH PROBLEM

Who Do You Trust With 5 Minutes Left?

How do coaches determine which players to trust in the final moments of a game?

goal: a data-driven framework for high-pressure lineup decisions

CLUTCH

- **Final 5 minutes of the 4th quarter or overtime**
- **Where score margin ⩽ 5 points at the 5-minute mark**

ABOUT OUR DATA

30 Years. 13.6 Million Moments. One Question.

PLAY-BY-PLAY

13.6M rows · 1996–present
Every game action logged
29,818 unique games

ALL-STAR SELECTIONS

2,058 rows · 1951–2026
Historical nominations
Career recognition proxy

PLAYER AWARD SHARES

3,465 rows · 1948–2025
MVP · DPOY · MIP · ROY
Award voting history

FINAL DATASET

2,771 players · 23 features · Clutch stats + Q1–Q3 predictors + Career signals
Minimum threshold: 100 clutch minutes / season)

SOURCE: NBA KAGGLE DATABASE

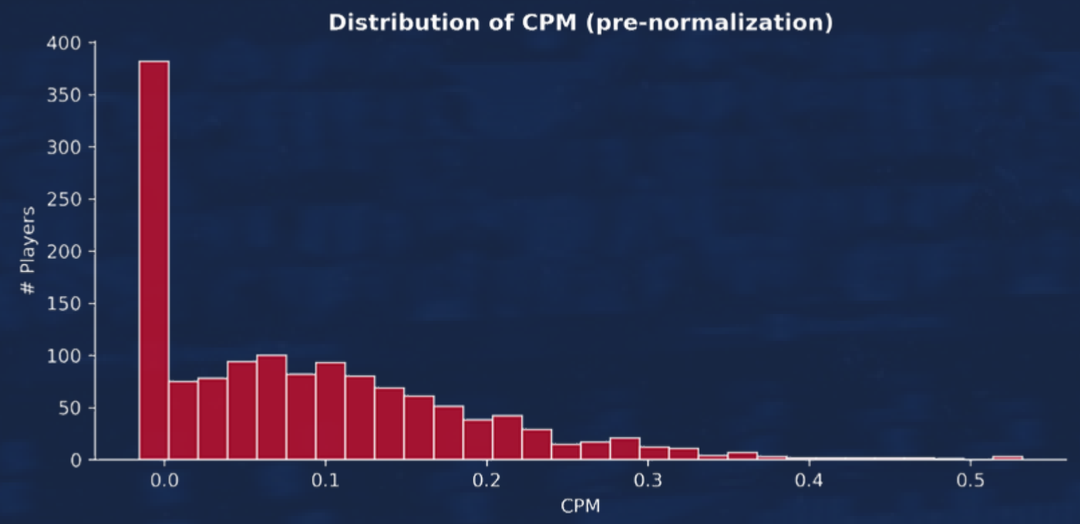
1,459,931
CLUTCH EVENTS

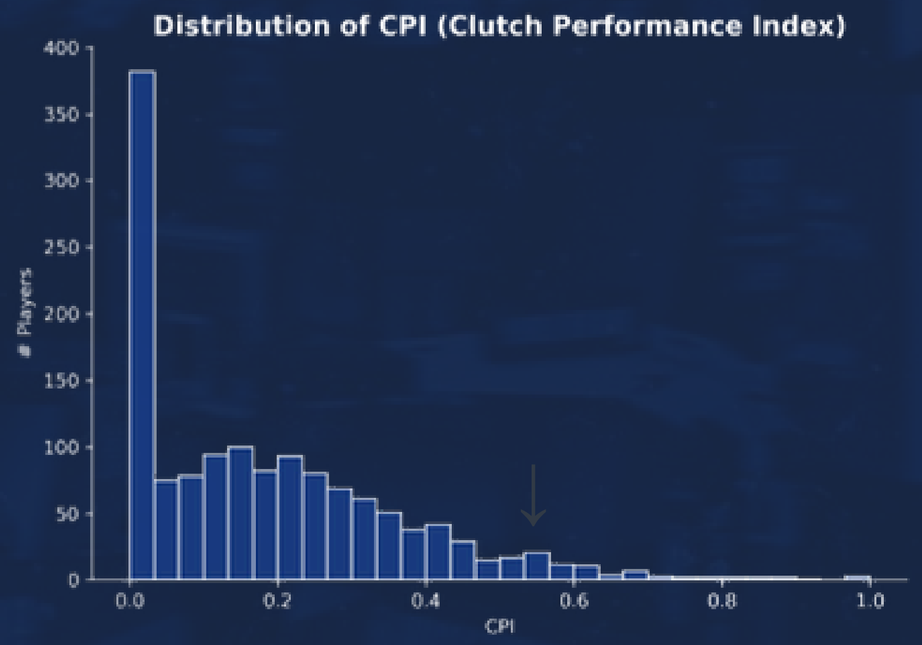
10.7%
OF ALL EVENTS

2771
PLAYERS

FROM COURT TO CALULATION

Defining Clutch





COMPOSITE PERFORMANCE METRIC (CPM)

$CPM = w_1(Scoring) + w_2(Efficiency) + w_3(Defense) - w_4(Turnovers)$

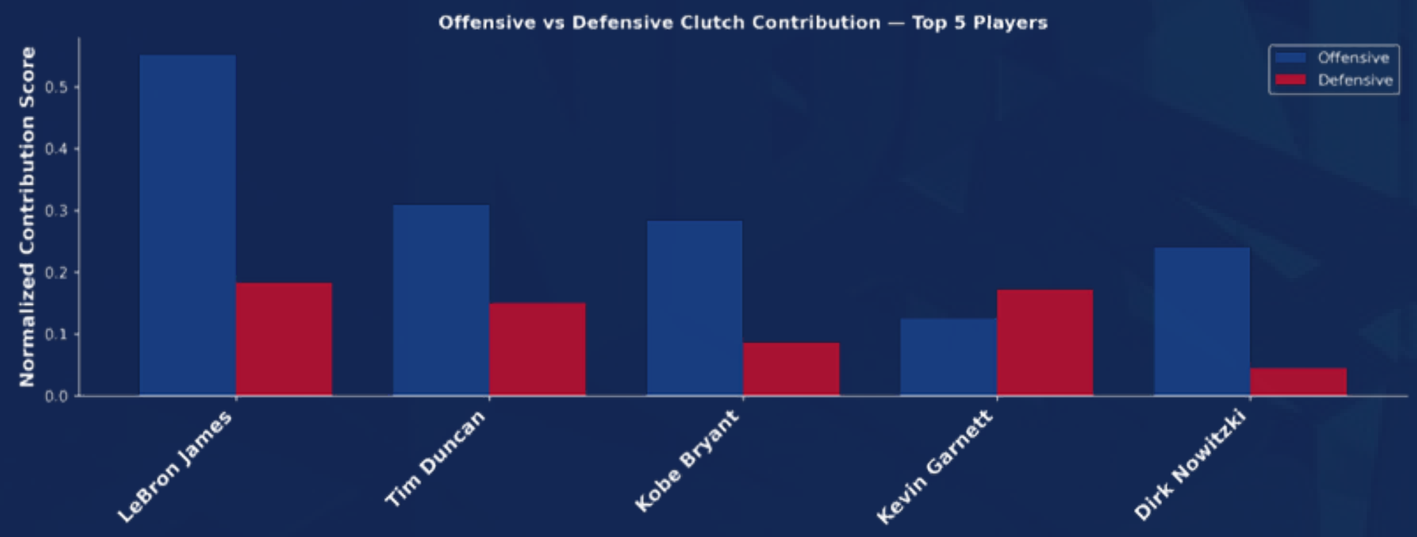
CLUTCH PERFORMANCE INDEX (CPI)

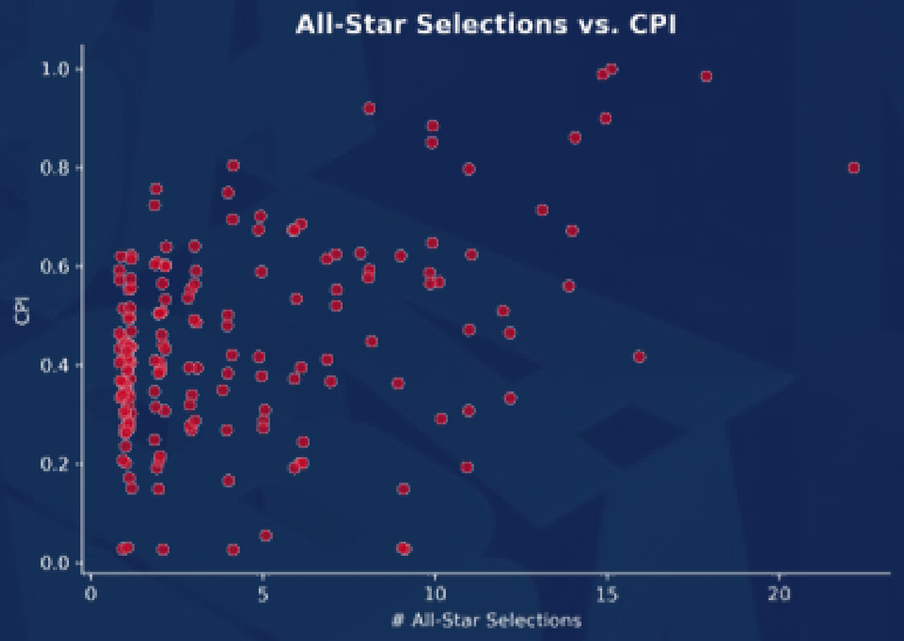
$$CPI = (CPM - min)/(max - min)$$

0 = worst clutch player

1 = best clutch player

DOES THE SCORE HOLD UP?





PREDICTING PRESSURE BEFORE IT ARRIVES

FEATURE SELECTION

Q1–Q3 performance stats + career signals as predictors. CPI as the target.

TRAIN / TEST SPLIT

80% for training, 20% held out.

MODEL RACE (5 FOLD CV)

Train the models and perfom 5-fold cross validation on train set

Evaluation

Evaluate on R2, RMSE, MAE

Feature Importance

Interpret results using Ridge coefficients

Residual Analysis

Conduct Residual Analysis

READING THE GAME BEFORE IT MATTERS

RIDGE REGRESSION

Finding linear patterns in player tendencies

CV R^2	0.827
Test R^2	0.792
RMSE	0.054
MAE	0.035

RANDOM FOREST

Hundreds of game situations

CV R^2	0.826

GRADIENT BOOSTING

Refines prediction round by round

CV R^2	0.805

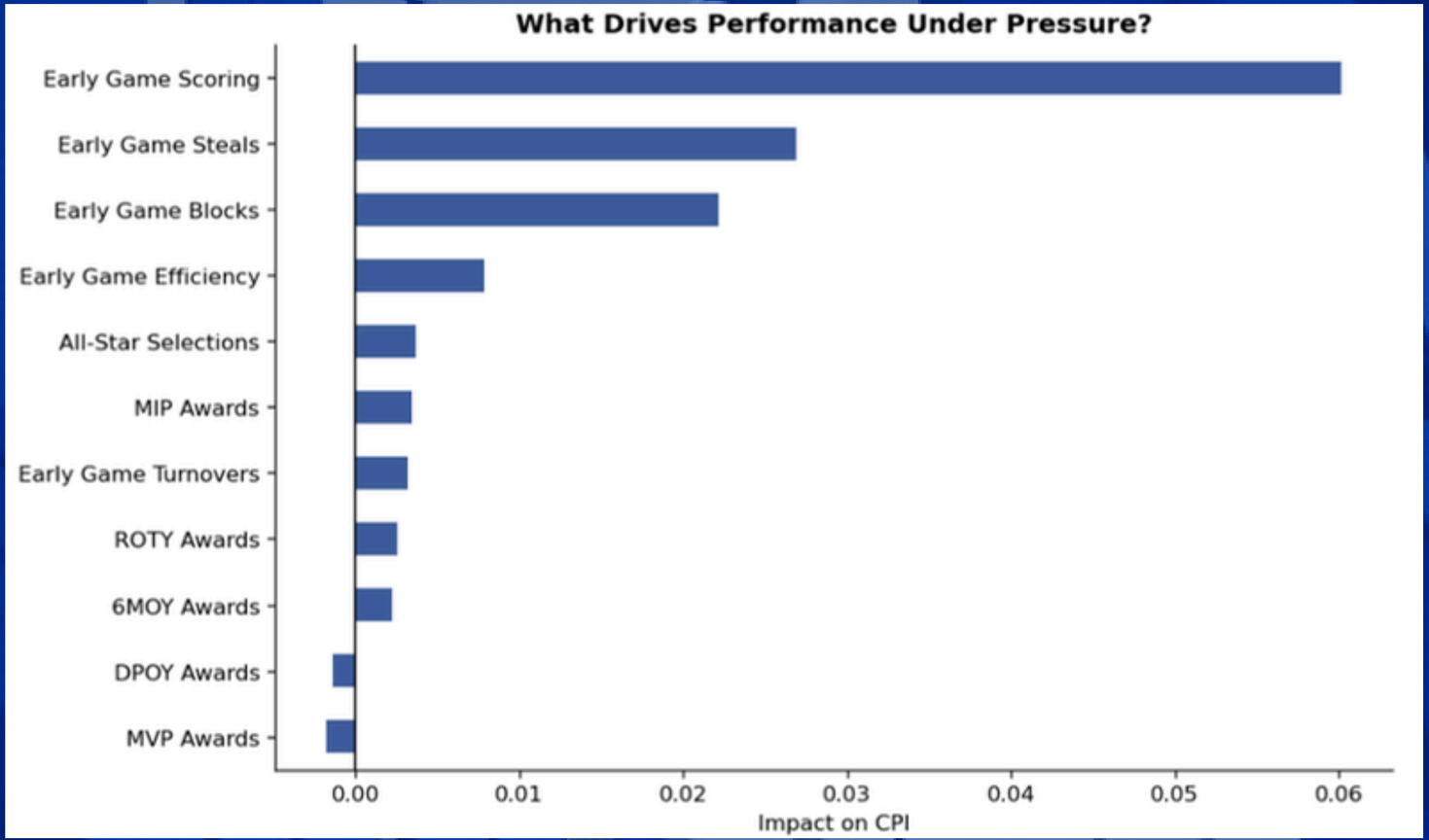
Ridge Regression was the best model; predicting a player's clutch performance index with 83% accuracy on training data and 79% on unseen players.

THE FOURTH QUARTER IS WON IN THE FIRST

Early game scoring is the strongest predictor of clutch performance

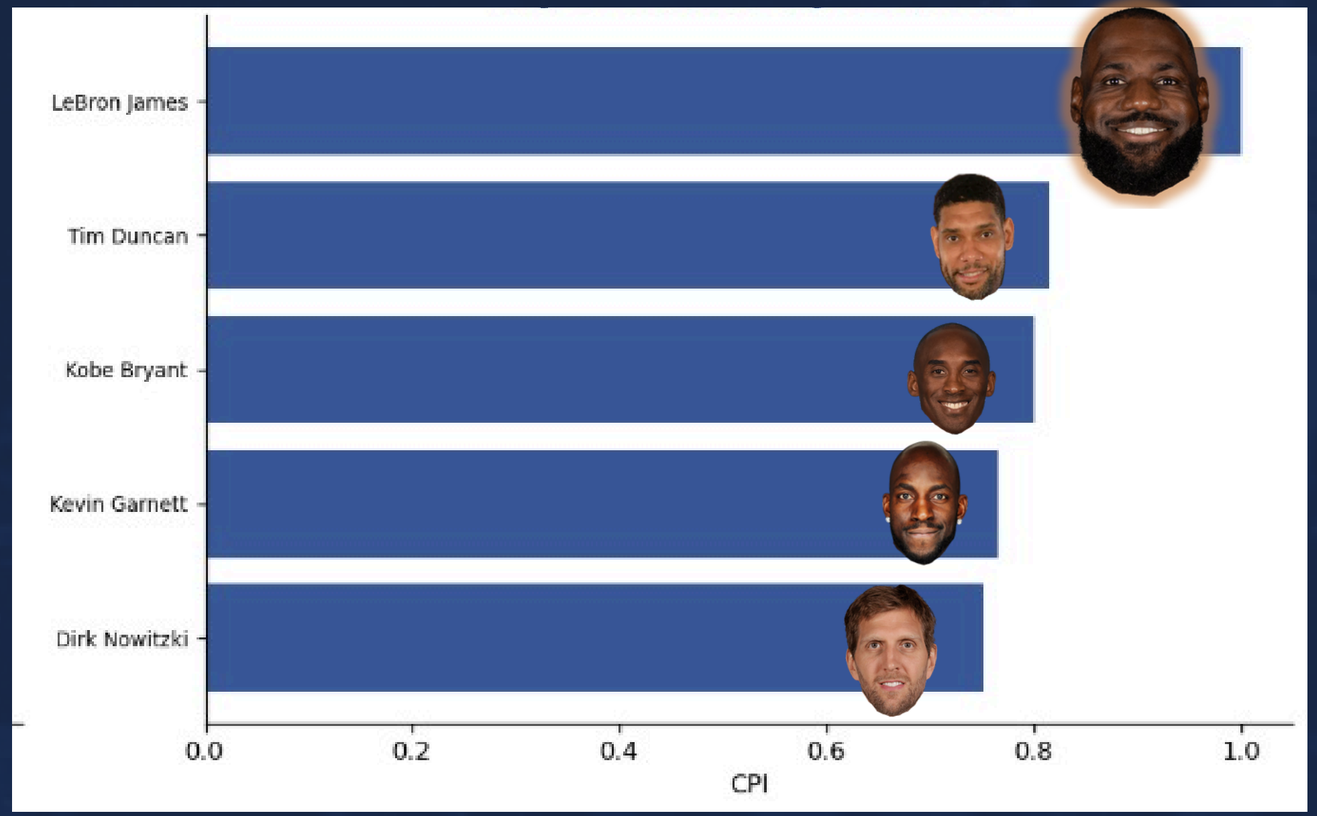
Defensive stats (steals, blocks) outrank every career accolade

Individual awards have near-zero predictive power



WHAT WE FOUND AT THE BUZZER

TOP 5 CLUTCH PLAYERS SINCE 1996



MVPs and All-Star selections show minimal impact

Clutch success is driven by in-game performance

Coaches should prioritize current game form over legacy metrics

BLIND SPOTS ON THE COURT

Does not account for defensive pressure or matchup difficulty

No adjustment for team context or play design

Sample size varies

THANK YOU